

Biodegradable Bio-based Material for Wearable Air Filtering Skin

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ABSTRACT

COVID-19, imagine having a temporary lip sticker that offers the protection of an n95 mask without the uncomfortable bulk. Using green electrospun nanofibers the lip sticker filters the virus and can communicate geospatial data to your phone using embedded NFC technology. Available in different designs and skins, some fiber formations can display temperature changes on your face. This paper investigates several prototypes of the described product.

CCS CONCEPTS

• **Human-centered Computing** → Human Computer Interaction (HCI); • **Interaction Styles**; • **Input Devices and Strategies**; • **User Interfaces**;

KEYWORDS

COVID, Mask, Temporary Tattoo, Electronic Skin, Nanofibers, Flexible Display, Materials Science, Electrospinning, Bio-based Material, Sustainability, Bio-HCI

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1 INTRODUCTION: MOTIVATION

Access to clean air was a dire environmental justice problem before the COVID-19 pandemic [1, 2]. Diseases caused by air pollutants (stroke, heart disease, lung cancer, etc.) were killing 7 million people a year and 9 out of 10 people were being exposed to tainted air daily [3]. Vulnerable, low-income and marginalized communities were bearing the brunt of disease caused by ubiquitous air pollution in their communities [4]. The pandemic compounded the clean air equity problem by targeting the same groups most impacted by pollution, the poor [5]. Those residing in high poverty areas were found to have the worst COVID-19 outcomes [5]. The government scrambled to find solutions that could slow the spread of disease and implemented measures such as mask mandates. With the widespread use of surgical masks, that carry only an efficacy of 75% when protecting against pathogens, a new unintended consequence

arose: an unrelenting generation of mask waste with the inability to degrade [6]. Studies found that there are more than 7000 tons of mask waste generated daily [7]. Creating a low – cost, biodegradable, wearable filtration device that helps vulnerable populations gain better access to clean air is the goal.

1.1 Cultural Foundations of Design

Ancient and indigenous face painting has been a form of technology for more than 10,000 years [8]. Across continents face painting has been used for ornamental decoration, traditional medicine, spirituality, skin protection, and war [9, 10]. This project draws inspirations from indigenous face painting designs and their role in protective healing to create wearable on skin air filtration technology.

1.2 Beyond Beauty Technology

Integrating skin and wearable computers has become a balance of art and functionality [11–15]. Wearables often serve as a beacon for expressive individuality [8]. However, device utility has slowly shifted towards granular practicality. Functions have moved from connecting with the outside world towards monitoring and benefiting what is happening inside the body [14, 16]. Wearable device utility has progressed from actuating music playback [9, 13], and opening doors [9], to strengthening joints with skin overlay [15], and regulating glucose and blood acidity levels [16]. This project connects users with the outside world while maintaining fashionable individuality and protecting the body from viral pollutant threats.

2 BASELINE FABRICATION

Electrospun soy protein isolate combined with poly-ethylene-oxide (PEO) was demonstrated to potentially filter at n95 levels by capturing 100% of aerosolized E. coli bacteria ranging in sizes from .3 micrometers – 5 micrometers in diameter [17]. N 95 masks are required to filter .3 micrometer (300nm) sized particles at 95 % [18]. Soy's low cost and biodegradability make it ideal for making short term air filtering materials. By incorporating herbal extracts the nanofibers gain added antiviral and antimicrobial abilities. All baseline fabrications were electrospun parameters from the E. coli paper.

Image A shows the most basic prototype that is fabricated with layers of gold leaf, electrospun soy and tattoo adhesive. The gold leaf makes the material potentially super conductive and enables for traces to be designed into the material in the form of a NFC tag. The traces could be applied to the lip layer as outlined in the Image B diagram or they could be incorporated into the chin as seen below in Image C. Incorporation of the NFC tag allows for the lip material to connect with phones. In further prototypes the NFC tag could be used to power sensors in the lip tattoo and give real time readings for bio or geospatial data. The NFC tags could also be

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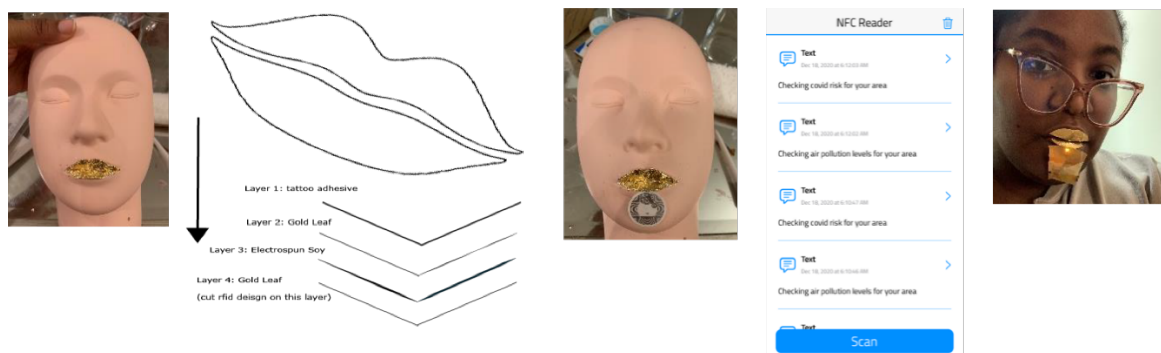


Figure 1: Image A shows a prototype for air filtration; Image B outlines the layers in fabrication; Image C illustrates how an NFC tag in device can be used to send geospatial data to the phone; Image D shows electrospun soy and gold leaf lip filter with light incorporated.

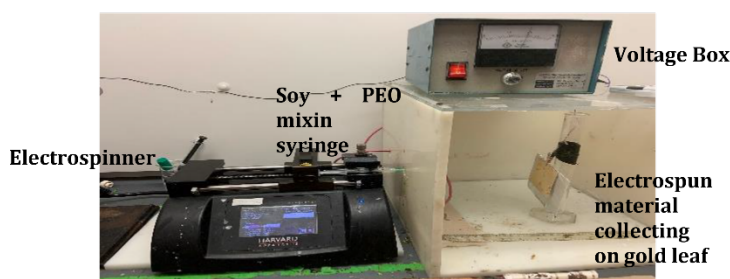


Figure 2: Electrospinning set up.

used to power UV -C LED lights on the face. Image D demonstrates what a light emitting lip filter could look like.

Figure 2 shows the lab setup for electrospinning soy protein isolate with PEO from the baseline fabrication. A syringe is placed in the electrospinning machine and the needle tip is connected to a voltage box. The machine pumps the mixture out and the electrical force creates charged threads of polymer. The threads collect on the gold leaf which is attached to an aluminum foil covered copper plate. The mixture was 70% soy and 30% PEO by weight.

2.1 Thermochromic Skin Fabrication

Instead of using gold leaf for layer 2 a thin silicon material was fabricated using embroidery stabilizer, tattoo adhesive and temporary tattoo silicone [17]. This thin material was attached to the aluminum foil covered copper plate and fibers were electrospun onto it. The initial soy mixture was modified to incorporate around 20% thermochromic pigment by weight (60 % soy and 20% PEO). The mixture was spun onto the thin silicon for 15 mins. After spinning the black pigment showed up in thin streaks across the silicon. To test whether the pigment effectively mixed with the soy and retained its properties a blow dryer was used. When the fibers are heated to over 77 ° F they change from black to a light pink. Future work will demonstrate how the silicon can be incorporated into layer 2. See Figure 3 for demonstration of the color change.



Figure 3: Thermochromic ink electrospun on silicon: left is below 77° F and right is heated by blow dryer to above 77° F

3 DISCUSSION

A low cost, biodegradable, highly conductive temporary air filtration device could be a game changer not only for those coping with the virus but also for people dealing with air pollution. Soy is a highly versatile material as demonstrated with the incorporation of thermochromic ink. Silver, copper or gold nanoparticles could be incorporated into the soy mixture to enhance filtration. Incorporating herbal extracts from common weeds have the ability to enhance antiviral and antibacterial function. Future work will incorporate more additives and actual testing of the materials. The fiber makeup will be compared with an N95 and documented using a scanning electron microscope (SEM).

The thermochromic filtration skin offers added utility. It presents a mechanism for altering the color, flexibility and durability of the prototype. Future work will demonstrate how it can be directly applied in the lip filtration device.

More work needs to be done to determine how NFC traces can be incorporated into the top layers of the lip filtration sticker. It may be possible to incorporate other power sources into the lip sticker beyond the radiowaves from the phone or battery power. Future iterations will look at prototyping the nose ring component of the lip filtration sticker. User studies will be conducted virtually and in person.

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REFERENCES

- [1] Wheeler, B. W., & Ben-Shlomo, Y. (2005). Environmental equity, air quality, socioeconomic status, and respiratory health: A linkage analysis of routine data from the health survey for England. *Journal of Epidemiology and Community Health*, 59(11), 948–954. doi:10.1136/jech.2005.036418
- [2] Lindgren, P., Johnson, J., Williams, A., Yawn, B., & Pratt, G. C. (2016). Asthma exacerbations and traffic: Examining relationships using link-based traffic metrics and a comprehensive patient database. *Environmental Health: A Global Access Science Source*, 15(1), 1–8. doi:10.1186/s12940-016-0184-2
- [3] World Health Organization. (2018). 9 out of 10 people worldwide breathe polluted air, but more countries are taking action. Geneva. Retrieved from: <https://www.who.int/news-room/detail/02-05-2018-9-out-of-10-people-worldwide-breathe-polluted-air-but-more-countries-are-taking-action>
- [4] Perera, F. P. (2017). Multiple threats to child health from fossil fuel combustion: Impacts of air pollution and climate change. *Environmental Health Perspectives*, 125(2), 141–148. doi:10.1289/EHP299
- [5] Little, C., Alsen, M., Barlow, J., Naymagon, L., Tremblay, D., Genden, E., Trosman, S., Iavicoli, L., & van Gerwen, M. (2021). The Impact of Socioeconomic Status on the Clinical Outcomes of COVID-19: a Retrospective Cohort Study. *Journal of Community Health: The Publication for Health Promotion and Disease Prevention*, 46(4), 794–802. <https://doi-org.proxy.library.cornell.edu/10.1007/s10900-020-00944-3>
- [6] Dharmaraj, S., Ashokkumar, V., Hariharan, S., Manibharathi, A., Show, P. L., Chong, C. T., & Ngamcharussrivichai, C. (2021). The COVID-19 pandemic face mask waste: A blooming threat to the marine environment. *Chemosphere*, 272. <https://doi-org.proxy.library.cornell.edu/10.1016/j.chemosphere.2021.129601>
- [7] Trafton, A. July 2021. **The environmental toll of disposable masks:** A new study calculates the waste generated by N95 usage and suggests possible ways to reduce it. MIT News. <https://news.mit.edu/2021/covid-masks-environment-0720>
- [8] Lise Eldridge. 2015. *Face paint: The story of makeup*. New York: Abrams Image.
- [9] Ruth Hopkins. 2020. The Evolving Use of War Paint in Native Communities. Retrieved from <https://www.teenvogue.com/story/use-war-paint-native-communities>
- [10] Tilahun Teklehaymanot and Mirutse Giday. 2010. Quantitative ethnobotany of medicinal plants used by Kara and Kwegu semi-pastoralist people in lower Omo River Valley, Debub Omo Zone, Southern Nations, Nationalities and Peoples Regional State, Ethiopia. *Journal of Ethnopharmacology*. <https://doi.org/10.1016/j.jep.2010.04.013>
- [11] Katia Vega and Hugo Fuks. 2013. Beauty Technology as an Interactive Computing Platform. ACM. New York, NY, USA, 357–360. DOI:<https://doi.org/10.1145/2512349.2512399>
- [12] Hsin-Liu Cindy Kao, Abdelkareem Bedri, and Kent Lyons. 2018. SkinWire: Fabricating a Self-Contained On-Skin PCB for the Hand. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 2, 3, Article 116 (September 2018), 23 pages. DOI:<https://doi.org/10.1145/3264926>
- [13] Hsin-Liu (Cindy) Kao, Christian Holz, Asta Roseway, Andres Calvo, and Chris Schmandt. 2016. DuoSkin: rapidly prototyping on-skin user interfaces using skin-friendly materials. In *Proceedings of the 2016 ACM International Symposium on Wearable Computers (ISWC '16)*. Association for Computing Machinery, New York, NY, USA, 16–23. DOI:<https://doi.org/10.1145/2971763.2971777>
- [14] Katia Vega, Nan Jiang, Xin Liu, Viirj Kan, Nick Barry, Pattie Maes, Ali K. Yetisan, and Joe Paradiso. 2018. The Dermal Abyss: Color-Changing Tattoos for Medical Diagnostics. *GetMobile: Mobile Comp. and Comm.* 22, 2 (June 2018), 38–42. DOI:<https://doi.org/10.1145/3276145.3276158>
- [15] Hsin-Liu (Cindy) Kao, Miren Bamforth, David Kim, and Chris Schmandt. 2018. Skinmorph: texture-tunable on-skin interface through thin, programmable gel. In *Proceedings of the 2018 ACM International Symposium on Wearable Computers (ISWC '18)*. Association for Computing Machinery, New York, NY, USA, 196–203. DOI:<https://doi.org/10.1145/3267242.3267262>
- [16] Amay J. Bandodkar, Wenzhao Jia, Ceren Yardımcı, Xuan Wang, Julian Ramirez, and Joseph Wang. *Analytical Chemistry* 2015 87 (1), 394–398 DOI: 10.1021/ac504300n
- [17] D. Lubasova, Anil Netravali, J. Parker, and B. Ingel. Bacterial filtration efficiency of green soy protein based nanofiber air filter. *J Nanosci Nanotechnol.* 2014 Jul;14(7):4891–8. doi: 10.1166/jnn.2014.8729. PMID: 24757959.
- [18] Qian Y, Willeke K, Grinshpun SA, Donnelly J, Coffey CC. Performance of N95 respirators: filtration efficiency for airborne microbial and inert particles. *Am Ind Hyg Assoc J.* 1998 Feb;59(2):128–32. doi: 10.1080/15428119891010389. PMID: 9487666.